

SOFTWARE REQUIREMENTS SPECIFICATION

Tool 1: Common OCR Converter for Handwritten Documents | Tool 2: GST and Income Tax Case Library System

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Date: 14 July 2026

1. Introduction

1.1 Purpose

This document describes, in plain language, the requirements for two new OCR (Optical Character Recognition) tools. It defines what each tool should do, who will use it, and what data can be used to train and test it. It is written so that a new intern with no prior background on the project can understand it and begin work immediately.

1.2 Background

The team has already built a Bank Statement Analyzer that uses OCR to read bank statements and extract useful data. That tool works well, but it only understands bank statement formats. This SRS defines the two tools that will be built next as part of this internship:

- Tool 1 – A general-purpose OCR converter that can read everyday handwritten documents other than bank statements.
- Tool 2 – A GST and Income Tax Case Library System to store, organise, search, and analyse case laws, departmental instructions, orders, and internal opinions related to Indian GST and Income Tax.

1.3 Scope

This SRS covers Tool 1 and Tool 2 only. The Bank Statement Analyzer is already built and is referenced here solely to describe how the existing OCR pipeline works.

1.4 Technology Note – OCR Is a Suggestion, Not a Rule

OCR is the default technology suggested for both tools because it is what the team already uses for the Bank Statement Analyzer. However, the development team is free to use a better-performing option if one delivers more accurate or faster results, for example:

- Cloud AI document APIs (e.g., Google Document AI, Azure AI Document Intelligence, AWS Textract).
- Vision-enabled AI models that can read images directly and handle messy handwriting better than plain OCR.
- Layout-aware models (e.g., LayoutLM, Donut) that understand both the text and its position on a form or receipt.

Whichever method is chosen, the tool's final output must still meet the requirements described in this document, including accuracy, structured data, and supported formats.

1.5 Intended Audience

- Interns and developers who will build the tools.
- QA / testers who will verify that the tools work correctly.
- Project manager who will track progress.

2. Tool 1 – Common OCR Converter for Handwritten Documents

2.1 Purpose

Build a general-purpose OCR tool that can read handwritten and printed documents other than bank statements. This converter is intended for everyday paperwork such as application forms, notes, letters, prescriptions, ID forms, exam sheets, or any other handwritten page a user might scan or photograph.

2.2 Description

A user uploads a photo, scan, or PDF of a handwritten or printed document. The tool reads the handwriting and text in the image and converts it into clean, editable digital text. The user can then copy, search, download, or export this text.

2.3 Functional Requirements

1. Accept input in the following formats: JPG, PNG, scanned image, and PDF.
2. Detect and read both handwritten and printed text on the page.
3. Handle multiple pages within one document (multi-page PDF or several images processed together).
4. Automatically correct tilted, rotated, or poorly lit images (auto-orientation and cleanup) before reading.
5. Display the extracted text alongside or over the original image, so the user can compare and correct mistakes.
6. Allow the user to manually edit or correct any incorrectly read word.
7. Allow the user to export the final text as TXT, DOCX, or CSV (if the document contains table-like data).
8. Show a confidence score for each line or word, so the user knows which parts may need manual review.
9. Maintain a history/log of previously processed documents for each user.
10. Support common handwriting styles (cursive and print) and common English/regional-language documents, with additional languages to be added later.

2.4 Non-Functional Requirements

- **Accuracy:** At least 85–90% correct reading on clear handwriting; the tool must flag lines it is unsure about.
- **Speed:** A single page should process in under 10 seconds.
- **Security:** Uploaded documents must be stored securely and visible only to the user who uploaded them.
- **Usability:** A first-time user should be able to upload a document and get results without training.
- **Scalability:** The tool should support many users uploading documents at the same time.

2.5 Intended Users

- Internal staff who need to digitize handwritten paperwork.
- Interns testing and improving the OCR model.
- Any team that currently retypes handwritten forms by hand.

2.6 Recommended Datasets for the Intern (Tool 1)

The following free, public datasets can be used to train and test the handwriting-recognition model:

Dataset	What It Contains	Good For
IAM Handwriting Database	Over 13,000 lines of handwritten English text from 657 different writers, including forms and word-level data.	Main dataset for training general handwriting recognition.
CVL Database	Handwritten pages in English and German from 310 writers.	Extra handwriting variety; testing the model on new writers.
MNIST	70,000 small images of handwritten digits (0–9).	Simple starting point for digit-recognition basics.
RIMES Database	French handwritten letters and forms.	Reference for building a similar dataset if regional languages are needed later.
Custom in-house samples	Scanned or photographed handwritten forms collected internally.	Making the model perform well on real-world, in-house documents.

3. Tool 2 – GST and Income Tax Case Library System

3.1 Introduction

The GST and Income Tax Case Library System is a web-based application to store, organise, search, and analyse case laws, departmental instructions, orders, and internal opinions related to Indian GST and Income Tax. It will be used by tax consultants, CA firms, in-house finance teams, and litigation teams to manage precedents, prepare replies and appeals, and track case outcomes more efficiently.

3.1.1 Purpose

- Define functional and non-functional requirements for developing the Case Library System.
- Provide a common reference for stakeholders: CA firm partners, tax consultants, developers, and testers.

3.1.2 Scope

The system will:

- Maintain a structured repository of GST and Income Tax case laws (ITAT, HC, SC, AAR, AAAR, Advance Rulings, departmental orders, internal notes).

- Allow powerful search by section, issue, court/forum, citation, keywords, parties, and assessment year, among other criteria.
- Track the case lifecycle for the firm's own matters (notices, replies, appeals, hearing dates, orders, and limitation dates).
- Generate standard reports for internal knowledge sharing, litigation management, and compliance tracking.

3.1.3 Definitions and Abbreviations

- GST: Goods and Services Tax.
- IT: Income Tax.
- ITAT: Income Tax Appellate Tribunal.
- AAR/AAAR: Authority / Appellate Authority for Advance Ruling.
- Case: Any judicial, administrative, or internal matter recorded in the system.
- Precedent: A case law or order used to support a legal argument.

3.2 Overall Description

3.2.1 Product Perspective

The system is a central knowledge and case-management application that may integrate with:

- Internal DMS (for PDFs of orders, replies, and similar documents).
- Email/calendar tools (for hearing-date reminders).
- External tax databases, through manual import or simple links only — not full integration in Phase 1.

3.2.2 User Classes and Characteristics

- Partner / Admin: Manages master data, user rights, and approval of new entries.
- Senior Consultant: Creates and validates case entries, tags issues, prepares notes, and uses advanced search.
- Junior Staff / Article: Searches and reads cases, drafts notes, and uploads documents (subject to approval).
- Client User (optional): Has view-only access to their own case status and key documents.

3.2.3 Operating Environment

- Web application accessible via standard browsers (Chrome, Edge, Firefox) on desktop and tablet.
- Backend: modern web stack (e.g., REST API, relational database) deployed on a secure cloud or on-premises server.

3.2.4 Design and Implementation Constraints

- Data must comply with Indian confidentiality and professional-conduct norms for client information.
- The system must support Indian date formats, tax terminology, and common case-law citation patterns.

3.2.5 Assumptions and Dependencies

- Users have basic IT literacy, including uploading documents and using search filters.
- Existing case documents (PDF/Word) are available for initial data migration.

3.3 System Features (Functional Requirements)

3.3.1 Case Law Repository

FR1: The user shall be able to create a new “Case Law” record with fields such as title, parties, forum (ITAT/HC/SC/AAR, etc.), section(s) involved, issue, assessment year, decision date, citation, and outcome (assessee-favourable / department-favourable / mixed).

FR2: The system shall allow attaching multiple documents (judgment PDF, summary, headnotes, internal note).

FR3: The system shall support tagging by GST/IT head, section, rule, notification, keyword, and subject (e.g., ITC, bogus purchases, reassessment).

FR4: The system shall maintain version history when a case record is edited (who changed what, and when).

3.3.2 Internal Case / Litigation Tracking

FR5: The user shall be able to create internal “Client Case” records covering notices, assessments, and appeals at different levels (AO, CIT(A), NFAC, ITAT, High Court, etc.).

FR6: Each case shall have client details, PAN/GSTIN, relevant year(s), officer details, key issues, monetary exposure, and status.

FR7: The system shall capture important dates: notice date, reply due date, hearing dates, order date, and appeal limitation last date.

FR8: The system shall allow linking a client case to relevant case laws in the repository (precedent mapping).

3.3.3 Advanced Search and Filters

FR9: Users shall be able to search case laws by any combination of forum, section, assessment year, issue, keyword, decision-date range, and outcome.

FR10: Search results shall support sorting (by date, forum, outcome, relevance) and pagination.

FR11: The system shall provide full-text search within user-uploaded PDFs where technically feasible (using OCR/indexing in a later phase).

3.3.4 Knowledge Notes and Templates

FR12: The user shall be able to add internal notes or “Practice Notes” summarising recurring issues, with links to key precedents.

FR13: The system shall allow storing standard templates (reply formats, appeal-grounds skeletons) mapped to issue types or forums.

3.3.5 Task, Diary, and Reminders

FR14: The system shall allow creating tasks against a case (e.g., draft reply, compile paper book, attend hearing).

FR15: The system shall send reminders (email/in-app) for approaching deadlines and hearing dates, configurable by the user (e.g., 7 days and 1 day before).

FR16: A calendar/diary view shall show all hearings and deadlines by user, client, and location.

3.3.6 Reporting and Dashboards

FR17: The system shall generate reports such as open cases by forum, exposure by client, and success ratio (favourable vs. unfavourable outcomes) by issue or officer.

FR18: The system shall provide a dashboard with key metrics: total active cases, upcoming hearings, ageing of open matters, and frequently used precedents.

3.3.7 User and Access Management

FR19: The system shall support role-based access control (Admin, Senior, Junior, Client).

FR20: Admin shall be able to restrict visibility of sensitive client cases and notes to specific users or groups.

FR21: The system shall maintain an audit log of logins and key actions (create/edit/delete) on cases and documents.

3.3.8 Data Import and Export

FR22: The system shall allow bulk import of case data from Excel/CSV templates for initial library creation.

FR23: Users shall be able to export filtered lists of cases to Excel/PDF for reporting to management or clients.

3.4 External Interface Requirements

3.4.1 User Interface

- Simple, responsive web UI with dashboards, filter panels, and tabular views of search results.
- Standardized data-entry forms with drop-down lists for forum, section, issue classification, and client selection to reduce errors.

3.4.2 Hardware Interfaces

- No special hardware is required beyond standard PCs/laptops, and optionally scanners for document digitization.

3.4.3 Software Interfaces

- Optional integration with email (SMTP) for outgoing reminders.
- Optional integration with calendar systems (ICS export/import) for hearings.

3.4.4 Communication Interfaces

- The application shall use HTTPS for all client-server communications.

3.5 Non-Functional Requirements

3.5.1 Performance

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- The system shall support at least [X] concurrent users without significant performance degradation.
 - Search queries on up to [Y] records should return results within [Z] seconds under normal conditions.

3.5.2 Security

- All data in transit shall be encrypted via HTTPS, and data at rest via database/server encryption, as per firm policy.
- User passwords shall be stored using secure hashing algorithms and session-management best practices.

3.5.3 Reliability and Availability

- Target uptime: at least 99% during working hours, excluding planned maintenance.
- Regular backups shall be taken, and recovery procedures documented, to avoid data loss.

3.5.4 Maintainability and Scalability

- The system shall be modular, with separate layers for UI, business logic, and data access.
- The design shall allow adding new tax types (e.g., customs, FEMA) and new forums without major rework.

3.5.5 Usability

- Navigation shall be intuitive, with minimal clicks from the dashboard to key functions such as search or case entry.
- Standard help text and tooltips shall be provided for key fields (e.g., outcome, issue tags).

3.6 Other Requirements

- Compliance with the firm's document-retention and confidentiality policies for client data and opinions.
- Logging suitable for internal audits on case handling and access to sensitive matters.

3.7 Appendices

- Appendix A: Sample data model (entities such as Client, Case, CaseLaw, Forum, IssueTag, Task, Document).
- Appendix B: Sample use cases and user stories (e.g., "Search for all ITC-related GST cases where the assessee won at ITAT in the last three years").

4. Suggested Next Steps for the Intern

11. Read this document fully and note down any doubts.
12. Download and explore one dataset per tool from the tables above.
13. Try a basic OCR engine first (e.g., Tesseract) to establish a baseline accuracy.
14. Compare results with one alternative option (e.g., a cloud OCR API) on the same sample images.